Advanced Topics in Regression

W. Evan Johnson, Ph.D.
Professor, Division of Infectious Disease
Director, Center for Data Science
Associate Director, Center for Biomedical Informatics and Health AI
Rutgers University – New Jersey Medical School

2026-10-28

Learning objectives

By the end of this lecture you should be able to:

- Define regression and its vocabulary (outcome, predictors, errors)
- Fit simple and multiple linear regression in R, and check residual plots
- Derive least squares as maximum likelihood under normal errors
- Compute estimates, standard errors, and tests by hand with matrices
- Recognize the regression fallacy (regression to the mean)
- Fit logistic and Poisson regression (GLMs) and interpret odds and rate ratios
- Recognize confounding and adjust for it

Introduction to Regression

We often want to know how two or more variables are related, e.g., how a child's height relates to the parents' heights.

Introduction to Regression

Regression estimates the relationship between:

- a **dependent variable**: outcome, response, or *label* in ML
- one or more **independent variables**: predictors, covariates, or *features* in ML

Introduction to Regression

The most common form is **linear regression**: find the line (or linear combination) that best fits the data under a chosen criterion.

Introduction to Regression

In math class, a line is (slope m, intercept b):

$$y = mx + b.$$

In statistics, we write:

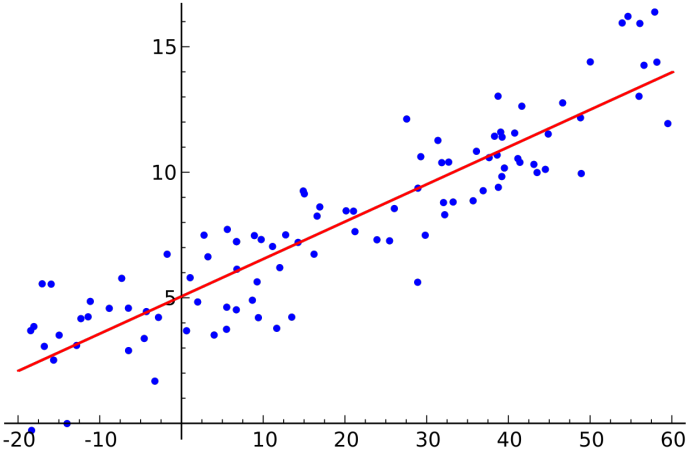
$$y_i = \beta_0 + x_i\beta_1,$$

with slope β_1, intercept β_0, and observed data points (x_i, y_i).

Introduction to Regression

- In math, lines fit the points *exactly*
- In statistics, points *do not* fall exactly on the line
- We find the **best fit** line to estimate the unknown slope and intercept: a **linear regression model**

Introduction to Regression



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Introduction to Regression

Formally, we observe n data pairs

$$(x_i, y_i), i = 1, ..., n,$$

and model

$$y_i = \beta_0 + x_i\beta_1 + \epsilon_i.$$

- ϵ_i are the **errors** (deviations from the line)
- β_0 and β_1 are unknown and must be *estimated* from the data

Introduction to Regression

Goal: find estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ that "best fit" the data.

Least squares: minimize the sum of squared residuals

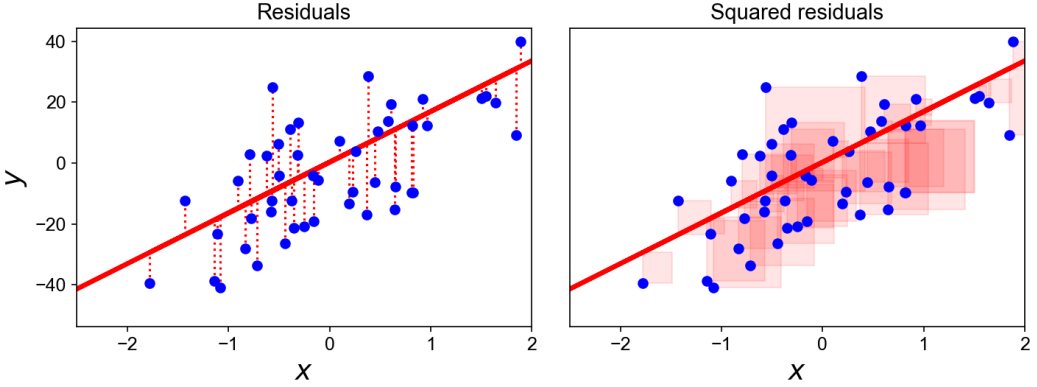
$$\min_{\beta_0,\beta_1} Q(\beta_0, \beta_1) = \inf_{\beta_0,\beta_1} \left\{ \sum_{i=1}^{n} (y_i - \beta_0 - x_i\beta_1)^2 \right\} = \inf_{\beta_0,\beta_1} \sum_{i=1}^{n} \epsilon_i^2.$$

Introduction to Regression

The least squares estimator (LSE) is the most common choice because of its properties:

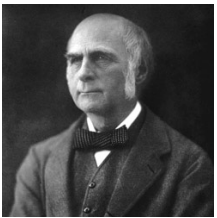
- It is the *best linear unbiased estimator* (Gauss-Markov)
- No other linear unbiased estimator has smaller sampling variance for $\hat{\beta}_0$ and $\hat{\beta}_1$

Introduction to regression



Case study: is height hereditary?

Francis Galton[1] studied heredity using family height data. Along the way, he developed **correlation** and **regression**.



[1] https://en.wikipedia.org/wiki/Francis_Galton

Case study: is height hereditary?

Galton's question: how well can we predict a child's height from the parents' heights? His answer: **regression**!

Historical note: Galton also promoted eugenics, a scientifically flawed movement with horrific consequences: https://pged.org/history-eugenics-and-genetics/.

Case study: is height hereditary?

Galton's data are in `HistData`. Keep each father and one randomly selected son:

```
library(tidyverse)
library(HistData)
data("GaltonFamilies")

set.seed(1983)
galton_heights <- GaltonFamilies %>%
  filter(gender == "male") %>%
  group_by(family) %>%
  sample_n(1) %>%
  ungroup() %>%
  select(father, childHeight) %>%
  rename(son = childHeight)
```

Case study: is height hereditary?

Both distributions are approximately normal, so summarize with means and SDs:

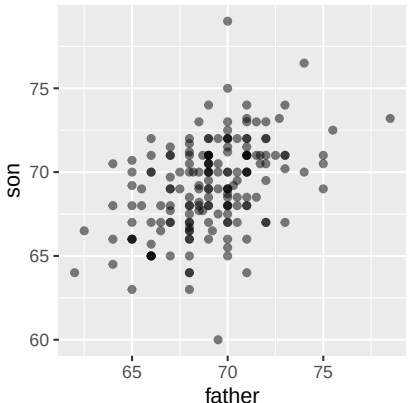
```
galton_heights %>%
  summarize(mean(father), sd(father), mean(son), sd(son))
```

```
## # A tibble: 1 x 4
##   `mean(father)` `sd(father)` `mean(son)` `sd(son)`
##            <dbl>        <dbl>       <dbl>     <dbl>
## 1           69.1         2.55        69.2      2.71
```

But this misses the key feature: taller fathers have taller sons.

Case study: is height hereditary?

```
galton_heights %>% ggplot(aes(father, son)) +
  geom_point(alpha = 0.5)
```



Case study: is height hereditary?

Correlation between father and son heights:

```
galton_heights %>%
    summarize(r = cor(father, son)) %>%
    pull(r)
```

```
## [1] 0.4334102
```

Case study: is height hereditary?

Fit the least squares model with `lm`:

$$y_i = \beta_0 + x_i\beta_1 + \varepsilon_i$$

(y_i = son's height, x_i = father's height)

```
fit <- lm(son ~ father, data = galton_heights)
fit$coef
```

```
## (Intercept)      father
##   37.287605    0.461392
```

Case study: is height hereditary?

summary gives more detail about the fit:

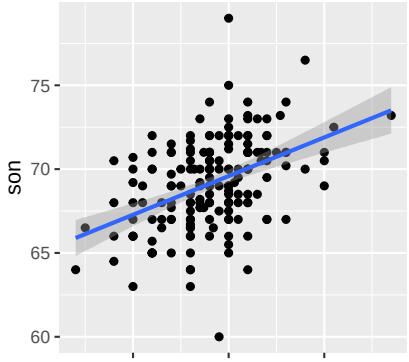
```
summary(fit)
```

```
##
## Call:
## lm(formula = son ~ father, data = galton_heights)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.3543 -1.5657 -0.0078  1.7263  9.4150
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 37.28761    4.98618   7.478 3.37e-12 ***
## father       0.46139    0.07211   6.398 1.36e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.45 on 177 degrees of freedom
## Multiple R-squared:  0.1878, Adjusted R-squared:  0.1833
```

Case study: is height hereditary?

Plot $\hat{Y}$ with confidence intervals in ggplot2:

```
galton_heights %>% ggplot(aes(father, son)) +
  geom_point() + geom_smooth(method = "lm")
```



Case study: is height hereditary?

`predict` returns predictions from an `lm` object, plus standard errors if requested:

```
fit <- galton_heights %>% lm(son ~ father, data = .)
y_hat <- fit %>% predict(se.fit = TRUE)
names(y_hat)
```

```
## [1] "fit"            "se.fit"         "df"             "residual.scale"
```

Introduction to Regression

Extend to p predictors:

$$y_i = x_{i1}\beta_1 + \ldots + x_{ip}\beta_p + \epsilon_i = \sum_{j=1}^{p} x_{ij}\beta_j + \epsilon_i.$$

To include an intercept, make one predictor constant ($x_{ij} = 1$ for all i), so $x_{ij}\beta_j = \beta_0$.

Introduction to Regression

Multiple regression on the full dataset:

```
fit2 <- GaltonFamilies %>%
  lm(childHeight ~ mother + father + gender + children, data = .)
fit2 %>% summary()
```

```
##
## Call:
## lm(formula = childHeight ~ mother + father + gender + children,
##     data = .)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.4759 -1.4743  0.0906  1.4789  9.1734
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 17.43103    2.77407   6.284 5.08e-10 ***
## mother       0.31619    0.03098  10.207  < 2e-16 ***
## father       0.38521    0.02898  13.292  < 2e-16 ***
## gendermale   5.19852    0.14197  36.617  < 2e-16 ***
## children    -0.04573    0.02631  -1.738   0.0825 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.162 on 929 degrees of freedom
## Multiple R-squared:  0.6366, Adjusted R-squared:  0.635
```

Introduction to Regression

Interpreting the model:

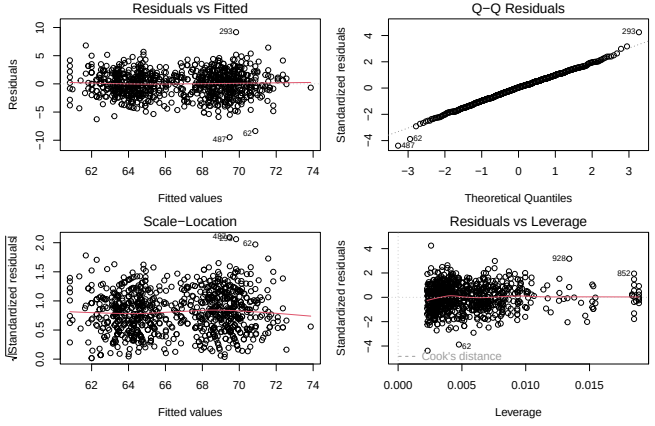
- Intercept: $\beta_0 = 17.4$ inches (p = 5.08e−10)
- Each inch of mother's height: $+0.32$ inches (p < 2e−16)
- Each inch of father's height: $+0.39$ inches (p < 2e−16)
- Male vs. female (reference): $+5.2$ inches (p < 2e−16). *R uses the first factor level (alphabetical) as the reference.*
- Number of children: not significant (p = 0.0825)
- $R^2 = 0.637$ (adjusted 0.635): `father`, `mother`, and `gender` are moderately strong predictors

Checking the model: residual diagnostics

Linear regression assumes linearity, constant variance, and roughly normal errors. Check with residual plots:

```
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1)); plot(fit2)
```

Checking the model: residual diagnostics



Checking the model: what to look for

- **Residuals vs. fitted:** a curve suggests a missing nonlinear term
- **Q-Q plot:** points off the line suggest non-normal errors (matters most for small samples)
- **Scale-location:** a funnel suggests non-constant variance (try a log transform)
- **Residuals vs. leverage:** points beyond Cook's distance lines are influential

Here everything looks fine.

Introduction to Regression

In matrix form, with outcome $\mathbf{y} = (y_1, \ldots, y_n)$, predictor matrix $\mathbf{X}$, and coefficients β $(p \times 1)$:

$$\mathbf{y} = X\beta + \epsilon, \quad \epsilon \sim N(\mathbf{0}, \sigma^2 I_n).$$

Introduction to Regression

Least squares then minimizes:

$$\hat{\beta} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - \sum_{j=1}^{p} x_{ij}\beta_j)^2 \right\} = \arg\min_{\beta} \{(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\}$$

Diversion: Maximum Likelihood for a Normal mean

Let $x_1, \ldots, x_N$ be independent Normal(μ, σ^2), with σ^2 known. The likelihood is

$$L(\mu|\mathbf{x}) = \left\{\frac{1}{2\pi\sigma^2}\right\}^{N/2} exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{N}(x_i - \mu)^2\right\}$$

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Maximizing $L(\mu|\mathbf{x})$ = minimizing $\sum_{i=1}^{N}(x_i - \mu)^2$ over μ.

Diversion: Maximum Likelihood for a Normal mean

Note the following:

$$\sum_{i=1}^{N}(x_i - \mu)^2 = \sum_{i=1}^{N}(x_i^2 - 2\mu x_i + \mu^2)$$

Diversion: Maximum Likelihood for a Normal mean

Note the following:

$$\begin{aligned}\sum_{i=1}^{N}(x_i-\mu)^2 &= \sum_{i=1}^{N}(x_i^2-2\mu x_i+\mu^2)\\ &= \sum_{i=1}^{N}x_i^2-2\mu\sum_{i=1}^{N}x_i+N\mu^2\end{aligned}$$

Diversion: Maximum Likelihood for a Normal mean

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Diversion: Maximum Likelihood for a Normal mean

therefore:

$$\frac{\partial}{\partial \mu} = -2 \sum x_i + 2N\hat{\mu} \stackrel{set}{=} 0$$

Diversion: Maximum Likelihood for a Normal mean

therefore:

$$\frac{\partial}{\partial \mu} = -2 \sum x_i + 2N\hat{\mu} \stackrel{set}{=} 0$$

$$\Rightarrow N\hat{\mu} = \sum x_i$$

Diversion: Maximum Likelihood for a Normal mean

therefore:

$$\begin{aligned}\frac{\partial}{\partial \mu} &= -2\sum x_i + 2N\hat{\mu} \overset{set}{=} 0 \\ &\Rightarrow N\hat{\mu} = \sum x_i \\ &\Rightarrow \hat{\mu} = \frac{\sum x_i}{N}\end{aligned}$$

Diversion: Maximum Likelihood for Regression

Recall the regression model:

$$\mathbf{y} = X\beta + \epsilon, \quad \epsilon \sim N(\mathbf{0}, \sigma^2 I_N).$$

Diversion: Maximum Likelihood for Regression

Extending the Normal mean MLE:

$$\sum_{i=1}^{N}(y_i - x_{i1}\beta_1 - \ldots - x_{ip}\beta_p)^2 = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta),$$

Diversion: Maximum Likelihood for Regression

Extending the Normal mean MLE:

$$\sum_{i=1}^{N}(y_i - x_{i1}\beta_1 - \ldots - x_{ip}\beta_p)^2 = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta),$$

so

$$L(\beta|\mathbf{X}, \mathbf{y}) = \left\{\frac{1}{2\pi\sigma^2}\right\}^{N/2} exp\left\{-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right\}.$$

Maximizing the likelihood = minimizing the sum of squared errors!

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

and therefore

$$\frac{\partial}{\partial\beta} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} \stackrel{set}{=} 0$$

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

and therefore

$$\begin{aligned}\frac{\partial}{\partial\beta} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} \overset{set}{=} 0 \\ &\Rightarrow \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}\end{aligned}$$

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

and therefore

$$\begin{aligned}
\frac{\partial}{\partial \beta} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} \stackrel{set}{=} 0 \\
&\Rightarrow \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y} \\
&\Rightarrow (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}
\end{aligned}$$

Diversion: Maximum Likelihood for Regression

Note

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta,$$

and therefore

$$\begin{aligned}
\frac{\partial}{\partial \beta} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} \stackrel{set}{=} 0 \\
&\Rightarrow \mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y} \\
&\Rightarrow (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\
&\Rightarrow \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}
\end{aligned}$$

Diversion: Maximum Likelihood for Regression

Now note:

$$E[\hat{\beta}] = E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}]$$

Diversion: Maximum Likelihood for Regression

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$$
\begin{aligned}
E[\hat{\beta}] &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{y}]
\end{aligned}
$$

Diversion: Maximum Likelihood for Regression

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$$
\begin{aligned}
E[\hat{\beta}] &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta
\end{aligned}
$$

Diversion: Maximum Likelihood for Regression

Now note:

$$
\begin{aligned}
E[\hat{\beta}] &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{y}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta \\
&= \beta
\end{aligned}
$$

Diversion: Maximum Likelihood for Regression

And, remembering that

$$Var[\mathbf{Ay}] = \mathbf{A}\{Var[\mathbf{y}]\}\mathbf{A}',$$

so therefore,

$$Var[\hat{\beta}] = Var[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}]$$

Diversion: Maximum Likelihood for Regression

And, remembering that

$$Var[\mathbf{Ay}] = \mathbf{A}\{Var[\mathbf{y}]\}\mathbf{A}',$$

so therefore,

$$\begin{aligned} Var[\hat{\beta}] &= Var[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\{Var[\mathbf{y}]\}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

Diversion: Maximum Likelihood for Regression

And, remembering that

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so therefore,

$$\begin{aligned} Var[\hat{\beta}] &= Var[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\{Var[\mathbf{y}]\}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\{\sigma^2 I_N\}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

Diversion: Maximum Likelihood for Regression

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Diversion: Maximum Likelihood for Regression

And, remembering that

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so therefore,

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Diversion: Maximum Likelihood for Regression

Test $H_0 : \beta_j = 0$ vs. $H_a : \beta_j \neq 0$ with:

$$t_{\beta_j} = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_{jj}}},$$

where v_{jj} is the jth diagonal of $\mathbf{V} = (\mathbf{X}'\mathbf{X})^{-1}$. Under H_0, $t_{\beta_j} \sim t_{N-p-1}$.

Case study: is height hereditary?

Using our estimator:

$$\widehat{\beta} = (X'X)^{-1}X'\mathbf{y}$$

```
X <- cbind(1,galton_heights$father)
y <- galton_heights$son
beta_hat <- solve(t(X)%*%X)%*%t(X)%*%y
beta_hat
```

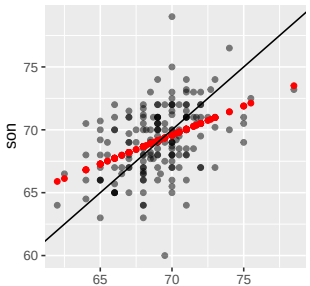
```
##             [,1]
## [1,] 37.287605
## [2,]  0.461392
```

Case study: is height hereditary?

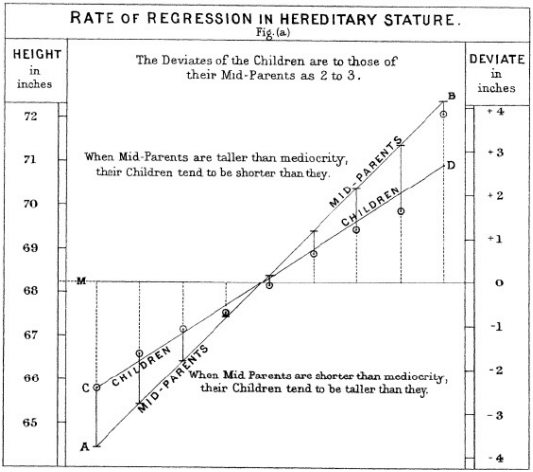
Predicted values: $\hat{y} = X\hat{\beta}$

```
y_hat <- X %*% beta_hat

galton_heights %>% ggplot(aes(father, son)) +
  geom_point(alpha = 0.5) +
  geom_point(aes(y = y_hat), col = "red") +
  geom_abline(slope = 1, intercept = 0)
```



Case study: is height hereditary?



Case study: is height hereditary?

Standard error for hypothesis testing:

$$SE(\hat{\beta}_i) = \hat{\sigma}\sqrt{v_{ii}},$$

where v_{ii} is the ith diagonal element of $(X'X)^{-1}$, and

$$\hat{\sigma}^2 = \frac{1}{N-p-1}(\mathbf{y} - X\widehat{\beta})'(\mathbf{y} - X\widehat{\beta})$$

```
N <- length(y)
p <- length(beta_hat)
sigma2_hat <- 1 / (N - p - 1) *
  t(y - X %*% beta_hat) %*% (y - X %*% beta_hat)
V <- solve(t(X) %*% X)
Z2 <- beta_hat[2] / sqrt(sigma2_hat * V[2, 2])
Z2
```

The Regression Fallacy

The *sophomore slump* (Wikipedia):

> *A second, or sophomore, effort fails to live up to the standards of the first: students, athletes (second season), bands (second album), TV shows, and movie sequels.*

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The Regression Fallacy

- MLB's Rookie of the Year (ROY) goes to the best first-year player
- ROY winners often do worse in year two: the "sophomore slump"
- Fox Sports[2]: "Will MLB's tremendous rookie class of 2015 suffer a sophomore slump?"

[2] http://www.foxsports.com/mlb/story/kris-bryant-carlos-correa-rookies-of-year-award-march-madness-francisco-lindor-kang-sano-120715

The Regression Fallacy

Is it real? Batting averages for the top ROYs do drop:

nameFirst	nameLast	rookie_year	rookie	sophomore
Willie	McCovey	1959	0.3541667	0.2384615
Ichiro	Suzuki	2001	0.3497110	0.3214838
Al	Bumbry	1973	0.3370787	0.2333333
Fred	Lynn	1975	0.3314394	0.3136095
Albert	Pujols	2001	0.3288136	0.3135593

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Proportion of ROYs with a lower sophomore average: 0.7053571.

Jitters or jinx? Look at all players with 130+ at bats (the ROY minimum) in both 2013 and 2014.

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Same pattern: most top performers drop.

nameFirst	nameLast	2013	2014
Miguel	Cabrera	0.3477477	0.3126023
Hanley	Ramirez	0.3453947	0.2828508
Michael	Cuddyer	0.3312883	0.3315789
Scooter	Gennett	0.3239437	0.2886364
Joe	Mauer	0.3235955	0.2769231

But these are not rookies!

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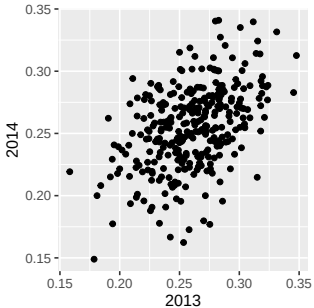
Now the worst performers of 2013:

nameFirst	nameLast	2013	2014
Danny	Espinosa	0.1582278	0.2192192
Dan	Uggla	0.1785714	0.1489362
Jeff	Mathis	0.1810345	0.2000000
B. J.	Upton	0.1841432	0.2080925
Adam	Rosales	0.1904762	0.2621951

Their batting averages mostly go up!

The Regression Fallacy

A reverse slump? No. There is no sophomore slump: year-to-year correlation is high, but not perfect.



The Regression Fallacy

Correlation is 0.460254 and the data look bivariate normal, so we predict 2014 average Y from 2013 average X with:

$$\frac{Y - .255}{.032} = 0.46 \left(\frac{X - .261}{.023} \right)$$

The Regression Fallacy

- Correlation < 1, so top 2013 performers are expected to do a bit worse in 2014
- Not a jinx, just chance
- ROYs are selected from the top of X, so Y will **regress to the mean**

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Generalized linear models (GLMs)

What if the outcome is binary (disease yes/no) or a count (number of infections)? A **GLM** has three parts:

- **Random component:** the outcome's distribution (normal, binomial, Poisson, ...)
- **Linear predictor:** $\eta_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip}$
- **Link function:** connects the mean to the linear predictor, $g(E[y_i]) = \eta_i$

Linear regression is the GLM with a normal outcome and identity link. GLMs are fit by maximum likelihood.

Common GLMs

Outcome	**Model**	**Link**
Continuous	Linear regression	identity: $E[y] = \eta$
Binary	Logistic regression	logit: $\log \frac{p}{1-p} = \eta$
Count	Poisson regression	log: $\log E[y] = \eta$

In R: `glm(y ~ x, family = binomial)` or `family = poisson`.

Logistic regression: diabetes in NHANES

NHANES (CDC National Health and Nutrition Examination Survey), adults 20+:

```
library(NHANES)
nh <- NHANES %>% filter(Age >= 20) %>% distinct(ID, .keep_all = TRUE)
fit_logit <- glm(Diabetes ~ Age + BMI + Gender, data = nh, family = binomial)
round(summary(fit_logit)$coefficients, 4)
```

```
##             Estimate Std. Error  z value Pr(>|z|)
## (Intercept)  -8.0227     0.3312 -24.2230    0e+00
## Age           0.0581     0.0032  18.0533    0e+00
## BMI           0.0892     0.0067  13.2254    0e+00
## Gendermale    0.3462     0.1000   3.4623    5e-04
```

Logistic regression: odds ratios

Coefficients are on the log-odds scale; exponentiate to get **odds ratios**:

```
round(exp(cbind(OR = coef(fit_logit), confint.default(fit_logit))), 3)
```

```
##                 OR 2.5 % 97.5 %
## (Intercept) 0.000 0.000  0.001
## Age         1.060 1.053  1.066
## BMI         1.093 1.079  1.108
## Gendermale  1.414 1.162  1.720
```

Each extra year of age multiplies the odds of diabetes by about 1.06; each BMI unit by about 1.093.

Poisson regression: counts

Number of live births for women 20+ in NHANES, by age and education:

```
nh_w <- nh %>% filter(Gender == "female", !is.na(nBabies), !is.na(Education))
fit_pois <- glm(nBabies ~ Age + Education, data = nh_w, family = poisson)
round(exp(cbind(RR = coef(fit_pois), confint.default(fit_pois))), 3)
```

```
##                              RR 2.5 % 97.5 %
## (Intercept)               2.129 1.844  2.458
## Age                       1.010 1.008  1.011
## Education9 - 11th Grade   0.813 0.717  0.922
## EducationHigh School      0.715 0.636  0.803
## EducationSome College     0.672 0.600  0.752
## EducationCollege Grad     0.637 0.565  0.718
```

Exponentiated coefficients are **rate ratios**: college graduates have 36% fewer births than the reference (8th grade or less), at the same age.

Poisson regression: check for overdispersion

Poisson assumes mean = variance. Check the ratio of residual deviance to its degrees of freedom:

```
fit_pois$deviance / fit_pois$df.residual
```

```
## [1] 0.5621043
```

- Well above 1: **overdispersion**; use `family = quasipoisson` or negative binomial (`MASS::glm.nb`) for honest standard errors
- Here it's about 0.56: births vary *less* than Poisson predicts, so these standard errors are conservative

Confounding

A **confounder** is related to both the predictor and the outcome, and can create, hide, or even reverse an association.

Classic example: UC Berkeley graduate admissions, 1973 (`UCBAdmissions` in R).

Confounding: Simpson's paradox

```
ucb <- as.data.frame(UCBAdmissions) %>%
  pivot_wider(names_from = Admit, values_from = Freq)
crude <- glm(cbind(Admitted, Rejected) ~ Gender, data = ucb,
             family = binomial)
adj   <- glm(cbind(Admitted, Rejected) ~ Gender + Dept, data = ucb,
             family = binomial)
round(exp(c(crude = coef(crude)["GenderFemale"],
            adjusted = coef(adj)["GenderFemale"])), 2)
```

```
##    crude.GenderFemale adjusted.GenderFemale
##                  0.54                  1.11
```

Confounding: Simpson's paradox

- Ignoring department: women had **lower** odds of admission (OR ≈ 0.54)
- Adjusting for department: slightly **higher** odds (OR ≈ 1.1)
- Women applied more to competitive departments: department confounds the comparison

Confounding in biomedical data

- Age, sex, smoking, socioeconomic status, and **batch** are common confounders
- Adjust by including confounders in the model (or by design: randomization, matching)
- Prediction vs. inference: confounders still help prediction, but you can't read coefficients as causal effects without thinking about confounding

Summary

- Linear regression finds the best-fitting line by least squares
- Least squares = maximum likelihood when errors are normal
- In matrix form: $\hat{\beta} = (X^T X)^{-1} X^T y$, with $Var(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$
- Galton's heights: regression was born studying heredity
- Regression to the mean explains the "sophomore slump": extreme performers drift back
- Check residual plots; GLMs extend regression to binary (logistic) and count (Poisson) outcomes
- Confounders can create or reverse associations; adjust for them

Session Info

```
sessionInfo()
```

```
## R version 4.3.3 (2024-02-29)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 24.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0
##
## locale:
## [1] C
##
## time zone: America/Denver
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] NHANES_2.1.4    Lahman_14.0-0   HistData_1.1.1  lubridate_1.9.3
##  [5] forcats_1.0.0   stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2
##  [9] readr_2.1.5     tidyr_1.3.1     tibble_3.2.1    ggplot2_3.4.4
## [13] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] utf8_1.2.4        generics_0.1.3    xml2_1.3.6        stringi_1.8.3
##  [5] lattice_0.22-5    hms_1.1.3         digest_0.6.34     magrittr_2.0.3
```